Selank

(Intranasal: 150–900 mcg/day speculative; 2.7 mg/day adjunct study, 14–21 days)

Selank is a synthetic tuftsin-derived heptapeptide developed in Russia for intranasal use. Its proposed effects involve GABAergic modulation, enkephalin metabolism, neurotrophic signaling, and immune regulation. [5,6,7,8,9,10,11] In small comparative clinical trials Selank reduced symptoms of anxiety with relatively little sedation; improved attention and stress resilience are plausible related benefits. [1,2,3]

💉 💉 Well tolerated during studied short courses; long-term risks remain incompletely characterized. [1,2,3,21]

⭐ ⭐ Probably effective for anxiety. [1,2,3]

⭐ Plausible cognitive support and stress resilience, especially when anxiety or stress impairs performance. [2,3,4,12,13,18,23]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Small clinical studies support relatively favorable short-course tolerability and less sedation than benzodiazepine comparators. [1,2,3] In the 2015 adjunctive study, reduced concentration impairment and fatigue were principally reductions in phenazepam-associated effects, not proof that Selank lacks adverse effects. [3] Rapid peptide degradation makes persistent accumulation less plausible, but does not establish long-term neurological, reproductive, immune, or organ safety. [22]

Dosing Patterns

Intranasal administration can produce acute central effects: a small healthy-volunteer study found changes in functional brain connectivity after administration, an imaging endpoint rather than a demonstration of better cognition. [4] Direct nose-to-brain transport through olfactory or trigeminal pathways is a plausible delivery mechanism. [9] Subjective onset within 15–60 minutes is anecdotal.

The following intranasal patterns are speculative experimental community patterns.

- **150–300 mcg per administration, one to three times daily; total 150–900 mcg daily**

An open prospective study randomly assigned 70 adults to phenazepam alone (30 participants) or phenazepam plus Selank 2.7 mg/day (40 participants) for 14 days; 20 of the latter continued Selank alone through day 21. This was an adjunctive protocol, not a dose-ranging monotherapy trial. [3] A reproduced Russian product instruction specifies two nasal drops in each nostril three times daily for 14 days, with a possible repeat course after 1–3 weeks following medical

consultation. Product-specific drop instructions should not be transferred to a different spray device. [22]

Benefits

- **Anxiety reduction**: Selank probably reduces anxiety in the studied anxiety/neurasthenia populations, with relatively little sedation in small active-comparator studies. [1,2]
- **Stress resilience**: a protective effect is plausible from clinical anxiety findings and animal stress research. [1,2,3,18,23]
- **Cognitive support**: improved attention and memory are plausible, particularly when anxiety, drug effects, or stress impair performance. [2,3,12,13]
- **Neurotrophic signaling**: Support for adaptive plasticity is plausible, rather than a uniform increase in BDNF. [8,13]
- **Immune modulation**: cytokine and immune-gene effects have been observed in human immune studies and animal experiments. Their clinical consequences depend on the tissue and challenge; they do not imply uniformly enhanced immunity. [5,18,20]
- **Antiviral activity**: activity against influenza A(H3N2) was observed in cell and mouse experiments, strongest with preventive administration. A corresponding human antiviral benefit is plausible. [14]

Community reports: calm, focus, and sleep benefits are anecdotal and should be distinguished from the measured outcomes above.

Potential Applications

- Anxiety disorders [1,2,3]
- Situational anxiety [1,2,23]
- Stress-related cognitive impairment [2,3,12,13,18]
- Post-stroke recovery [19,20]

Cautions

- **Pregnancy, breastfeeding, and children**: the Russian product instruction contraindicates use during pregnancy or breastfeeding and under age 18. [22]
- **Unstable psychiatric conditions**: caution is appropriate in psychosis or bipolar mania. Major psychiatric comorbidities were excluded from the 2015 study, so its tolerability findings should not be generalized to these populations. [3]
- **CNS depressants and psychoactive polypharmacy**: interactions may be context dependent. Selank altered olanzapine-associated gene responses in cultured cells and modified benzodiazepine effects in animal and small human studies. These findings support caution, not a predictable interaction magnitude for an individual combination. [3,10,11,23]

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Side Effects

- **Sleep effects**: sleep disturbance or vivid dreams are anecdotal reports, particularly with late dosing.
- **Activation**: transient anxiety or activation is reported anecdotally; distinguish this from the generally favorable short-course tolerability of the studied populations. [1,2,3]

Biological Mechanism

Selank combines the tuftsin sequence Thr-Lys-Pro-Arg with Pro-Gly-Pro and has multimodal biological effects. [9,23] Binding experiments support a proposed allosteric influence on GABAergic signaling. Rat frontal-cortex transcriptional changes partly overlapped those induced by GABA, but the pooled-tissue design limits assessment of between-animal variability. In IMR-32 cells, Selank alone did not significantly alter the measured genes; its combinations with GABA or olanzapine produced different responses. These findings favor context-dependent modulation rather than a simple, uniformly stimulatory GABA mechanism. [9,10,11] Inhibition of plasma enkephalin degradation provides a separate plausible mechanism, with phenotype-dependent behavioral/enzyme effects in mice and reduced morphine-withdrawal signs in rats. [6,7,16,17] Rat learning and ethanol-impairment experiments also implicate monoamine metabolism and context-dependent BDNF regulation. [8,12,13] Human cytokine observations and animal stress/ischemia transcriptomics support immune modulation. [5,18,20] Together, these pathways plausibly influence plasticity, neuroprotection, and anxiety/stress responses; their contributions need not be identical across physiological states.

Combining Selank and Semax

Combining Selank and Semax is a speculative community practice. Their partly distinct but overlapping central effects make complementary effects on anxiety, stress resilience, attention, and learning plausible. The healthy-volunteer connectivity study examined the peptides separately. [4] Additive activation, sleep disturbance, or mood effects are plausible. Specific combined doses, sequencing strategies, alternating-day schedules, and cycling patterns remain speculative.

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References

[1] Zozulia AA, Neznamov GG, Siuniakov TS, et al. [Efficacy and possible mechanisms of action of a new peptide anxiolytic selank in the therapy of generalized anxiety disorders and neurasthenia]. *Zh Nevrol Psikhiatr Im S S Korsakova*. 2008;108(4):38-48. https://pubmed.ncbi.nlm.nih.gov/18454096/

[2] Medvedev VE, Tereshchenko ON, Israelian AIu, et al. [A comparison of the anxiolytic effect and tolerability of selank and phenazepam in the treatment of anxiety disorders]. *Zh Nevrol Psikhiatr Im S S Korsakova*. 2014;114(7):17-22. https://pubmed.ncbi.nlm.nih.gov/25176261/

[3] Medvedev VE, Tereshchenko ON, Kost NV, et al. [Optimization of the treatment of anxiety disorders with selank]. *Zh Nevrol Psikhiatr Im S S Korsakova*. 2015;115(6):33-40. https://doi.org/10.17116/jnevro20151156133-40

[4] Panikratova YR, Lebedeva IS, Sokolov OY, et al. Functional Connectomic Approach to Studying Selank and Semax Effects. *Dokl Biol Sci*. 2020;490(1):9-11. https://doi.org/10.1134/S001249662001007X

[5] Uchakina ON, Uchakin PN, Miasoedov NF, et al. [Immunomodulatory effects of selank in patients with anxiety-asthenic disorders]. *Zh Nevrol Psikhiatr Im S S Korsakova*. 2008;108(5):71-75. https://pubmed.ncbi.nlm.nih.gov/18577961/

[6] Zozulya AA, Kost NV, Sokolov OY, et al. The inhibitory effect of Selank on enkephalin-degrading enzymes as a possible mechanism of its anxiolytic activity. *Bull Exp Biol Med*. 2001;131(4):315-317. https://doi.org/10.1023/A:1017979514274

[7] Zolotarev YA, Sokolov OY, Kost NV, Vaskovsky BV, Myasoedov NF, Zozulya AA. Leu-Enkephalin Generally Labeled with Tritium in Studying the Selank Inhibiting Effect on the Enkephalin-Degrading Enzymes of Human Blood Plasma. *Russ J Bioorg Chem*. 2004;30(3):207-212. https://doi.org/10.1023/B:RUBI.0000030126.09208.c3

[8] Inozemtseva LS, Karpenko EA, Dolotov OV, et al. Intranasal administration of the peptide Selank regulates BDNF expression in the rat hippocampus in vivo. *Dokl Biol Sci*. 2008;421:241-243. https://doi.org/10.1134/S0012496608040066

[9] Volkova A, Shadrina M, Kolomin T, et al. Selank Administration Affects the Expression of Some Genes Involved in GABAergic Neurotransmission. *Front Pharmacol*. 2016;7:31. https://doi.org/10.3389/fphar.2016.00031

[10] Vyunova TV, Andreeva L, Shevchenko K, Myasoedov N. Peptide-based Anxiolytics: The Molecular Aspects of Heptapeptide Selank Biological Activity. *Protein Pept Lett*. 2018;25(10):914-923. https://doi.org/10.2174/0929866525666180925144642

[11] Filatova E, Kasian A, Kolomin T, et al. GABA, Selank, and Olanzapine Affect the Expression of Genes Involved in GABAergic Neurotransmission in IMR-32 Cells. *Front Pharmacol*. 2017;8:89. https://doi.org/10.3389/fphar.2017.00089

[12] Semenova TP, Kozlovskiĭ II, Zakharova NM, Kozlovskaia MM. [Experimental optimization of learning and memory processes by selank]. *Eksp Klin Farmakol*. 2010;73(8):2-5. https://pubmed.ncbi.nlm.nih.gov/20919548/

[13] Kolik LG, Nadorova AV, Antipova TA, Kruglov SV, Kudrin VS, Durnev AD. Selank, Peptide Analogue of Tuftsin, Protects Against Ethanol-Induced Memory Impairment by Regulating of BDNF Content in the Hippocampus and Prefrontal Cortex in Rats. *Bull Exp Biol Med*. 2019;167(5):641-644. https://doi.org/10.1007/s10517-019-04588-9

[14] Ershov FI, Uchakin PN, Uchakina ON, Mezentseva MV, Alekseeva LA, Miasoedov NF. [Antiviral activity of immunomodulator Selank in experimental influenza infection]. *Vopr Virusol*. 2009;54(5):19-24. https://pubmed.ncbi.nlm.nih.gov/19882898/

[15] Yasenyavskaya AL, Samotrueva MA, Tsibizova AA, Bashkina OA, Myasoedov NF, Andreeva LA. The Influence of Selank on the Level of Cytokines Under the Conditions of "Social" Stress. *Curr Rev Clin Exp Pharmacol*. 2021;16(2):162-167. https://doi.org/10.2174/1574884715666200704152810

[16] Konstantinopolsky MA, Chernyakova IV, Kolik LG. Selank, a Peptide Analog of Tuftsin, Attenuates Aversive Signs of Morphine Withdrawal in Rats. *Bull Exp Biol Med*. 2022;173(6):730-733. https://doi.org/10.1007/s10517-022-05624-x

[17] Sokolov OY, Meshavkin VK, Kost NV, Zozulya AA. Effects of Selank on behavioral reactions and activities of plasma enkephalin-degrading enzymes in mice with different phenotypes of emotional and stress reactions. *Bull Exp Biol Med*. 2002;133(2):133-135. https://doi.org/10.1023/A:1015582302311

[18] Filippenkov IB, Glazova NY, Sebentsova EA, et al. Selank Peptide Causes Changes in Gene Expression in the Hippocampus of Rats in the Early Hours after Acute Restraint Stress. *Nanobiotechnol Rep*. 2024;19(3):381-392. https://doi.org/10.1134/S2635167624601335

[19] Skvortsova VI, Tvorogova TV, Dubina AI, et al. [Experimental hemorrhagic stroke: the study of neuropeptides (MIF, selank) in the intraperitoneal injection]. *Zh Nevrol Psikhiatr Im S S Korsakova*. 2009;109(12 Suppl 2):62-66. https://pubmed.ncbi.nlm.nih.gov/20873405/

[20] Filippenkov IB, Shpetko YY, Stavchansky VV, et al. Transcriptomic Effects of Peptide Drug Selank during Early Experimental Cerebral Ischemia in Rats. *Mol Biol*. 2026;60:530-542. https://doi.org/10.1134/S002689332670007X

[21] US Food and Drug Administration. Certain bulk drug substances for use in compounding that may present significant safety risks: Selank acetate. Accessed October 26, 2026. https://www.fda.gov/drugs/human-drug-compounding/certain-bulk-drug-substances-use-compounding-may-present-significant-safety-risks

[22] Peptogen. Selank nasal drops: instruction for medical use. Russian product information reproduced by GEOTAR. Accessed October 26, 2026. https://files.dev.lsgeotar.ru/public/qrKcpEfh8H8FnO7ar2GSoQ.pdf

[23] Kasian A, Kolomin T, Andreeva L, et al. Peptide Selank Enhances the Effect of Diazepam in Reducing Anxiety in Unpredictable Chronic Mild Stress Conditions in Rats. *Behav Neurol*. 2017;2017:5091027. https://doi.org/10.1155/2017/5091027